Test 9 - Answer Key

Subject: Computer Networks

Total Marks: 50 | Time: 120 minutes

Q1. Name one application-layer protocol that commonly uses UDP as its transport protocol. [1 marks | Topic: Transport Layer | Bloom: Understand]

Answer: DNS (Domain Name System) or DHCP (Dynamic Host Configuration Protocol) or RTP (Real-time Transport Protocol for streaming).

Q2. State one primary function of the Network Layer in the TCP/IP model. [1 marks | Topic: Network Layer | Bloom: Understand]

Answer: Routing / Logical addressing / Inter-network communication / Packet forwarding between different networks.

Q3. What is the main function of the network layer? [2 marks | Topic: Network Layer | Bloom: Remember]

Answer: The network layer is responsible for logical addressing, routing, and forwarding packets from source to destination across different networks.

Q4. Explain how port numbers facilitate multiplexing and demultiplexing at the Transport Layer. [2 marks | Topic: Transport Layer | Bloom: Understand]

Answer: Port numbers are used by the Transport Layer to identify specific application processes running on a host. For demultiplexing, when a segment arrives from the network, the transport layer uses the destination port number to deliver the data to the correct application process. For multiplexing, the transport layer adds a source port number to each segment from an application, allowing multiple applications to share the same network connection.

Q5. Describe two primary functions of the Network Layer in the TCP/IP model, providing a brief explanation for each. [4 marks | Topic: Network Layer | Bloom: Understand]

Answer: Two primary functions of the Network Layer are:

1. **Logical Addressing (IP Addressing):** The Network Layer assigns unique logical addresses (IP addresses) to devices. These addresses are used to uniquely identify devices across different networks, enabling global communication independent of the underlying physical network hardware.

2. **Routing/Forwarding:** This layer is responsible for determining the best path for data packets to travel from the source host to the destination host across multiple interconnected networks. Routers at this layer use routing tables to make decisions on where to forward packets based on their destination IP address.

Q6. Analyze what happens if a router does not have a route for a destination [5 marks | Topic: Network Layer | Bloom: Analyze]

Answer: If no route exists, the router drops the packet. It may send an ICMP destination unreachable message to the source, depending on configuration.

Q7. Evaluate whether TCP or UDP is more suitable for file transfer. Justify your answer. [5 marks | Topic: Transport Layer | Bloom: Evaluate]

Answer: TCP is more suitable for file transfer because reliability and correct ordering are important. TCP ensures lost data is retransmitted and delivered in order, while UDP does not provide such guarantees.

Q8. A company has been allocated the Class B network address 172.16.0.0/16. They need to divide this into at least 5 subnets, with each subnet accommodating a minimum of 500 hosts. Design the subnetting scheme. For the first two usable subnets, provide the following:

a) Network Address

b) Subnet Mask (in CIDR notation and dotted decimal)

c) First Usable Host IP Address

d) Last Usable Host IP Address

e) Broadcast Address

f) Maximum number of hosts supported by each subnet. [10 marks | Topic: Network Layer | Bloom: Apply]

Answer: To accommodate at least 500 hosts per subnet, we need 9 host bits (2^9 - 2 = 510 usable hosts). The original /16 network mask needs to be extended to 32 - 9 = 23 bits. So, the new subnet mask is /23 (255.255.254.0).

Subnet 1:

a) Network Address: 172.16.0.0

b) Subnet Mask: /23 (255.255.254.0)

c) First Usable Host IP: 172.16.0.1

d) Last Usable Host IP: 172.16.1.254

e) Broadcast Address: 172.16.1.255

f) Maximum Hosts: 510

Subnet 2:

a) Network Address: 172.16.2.0

b) Subnet Mask: /23 (255.255.254.0)

c) First Usable Host IP: 172.16.2.1

d) Last Usable Host IP: 172.16.3.254

e) Broadcast Address: 172.16.3.255

f) Maximum Hosts: 510

Q9. A router has the following simplified routing table entries:

Destination Network	Next Hop	Interface
192.168.10.0/24	10.0.0.1	Eth0
192.168.20.0/24	10.0.0.2	Eth1
192.168.30.0/24	10.0.0.3	Eth2
0.0.0.0/0 (Default)	10.0.0.4	Eth3

For each of the following destination IP addresses, determine the Next Hop IP address and the outgoing Interface that the router will use to forward the packet. Justify your answer based on the longest prefix match rule.

a) 192.168.10.15

b) 192.168.20.100

c) 192.168.40.5

d) 192.168.30.250

e) 192.168.10.250 [10 marks | Topic: Network Layer | Bloom: Apply]

Answer: a) Destination: 192.168.10.15

Next Hop: 10.0.0.1, Interface: Eth0

Justification: The destination 192.168.10.15 matches 192.168.10.0/24 (prefix length 24) and 0.0.0.0/0 (prefix length 0). 192.168.10.0/24 is the longest prefix match.

b) Destination: 192.168.20.100

Next Hop: 10.0.0.2, Interface: Eth1

Justification: The destination 192.168.20.100 matches 192.168.20.0/24 (prefix length 24) and 0.0.0.0/0. 192.168.20.0/24 is the longest prefix match.

c) Destination: 192.168.40.5

Next Hop: 10.0.0.4, Interface: Eth3

Justification: The destination 192.168.40.5 only matches the default route 0.0.0.0/0 (prefix length 0), as there's no more specific entry.

d) Destination: 192.168.30.250

Next Hop: 10.0.0.3, Interface: Eth2

Justification: The destination 192.168.30.250 matches 192.168.30.0/24 (prefix length 24) and 0.0.0.0/0. 192.168.30.0/24 is the longest prefix match.

e) Destination: 192.168.10.250

Next Hop: 10.0.0.1, Interface: Eth0

Justification: The destination 192.168.10.250 matches 192.168.10.0/24 (prefix length 24) and 0.0.0.0/0. 192.168.10.0/24 is the longest prefix match.

Q10. Describe the purpose of flow control in the Transport Layer and explain how TCP implements this using the sliding window mechanism. [10 marks | Topic: Transport Layer | Bloom: Understand]

Answer: Flow control is a mechanism in the Transport Layer designed to prevent a fast sender from overwhelming a slow receiver by sending data faster than the receiver can process or store it. It ensures that the receiver's buffer does not overflow, leading to data loss.